MIL-B-24155B(SH)
27 October 1986
SUPERSEDING
MIL-B-24155A(SHIPS)
18 May 1967
(See 6.5)

MILITARY SPECIFICATION

BOILER SCALE REMOVING COMPOUND

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a dual packaged compound to be used as a scale remover on medium and high pressure boilers. One package shall contain a uniform mixture of crystalline sulfamic acid, anhydrous citric acid, and microfine silica. The other package shall contain the acid inhibitor, diethylthiourea.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL
PPP-D-723 - Drums, Fiber.

STANDARDS

FEDERAL
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 6850
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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DEPARTMENT OF LABOR (OSHA)

Code of Federal Regulations (CFR)
29 CFR, Part 1910.1200 - Hazard Communication Standard.

(The Code of Federal Regulations (CFR) is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. The nominal composition based on the use of commercially available chemicals shall be 93.7 percent sulfamic acid and 6.3 percent anhydrous citric acid in the acid mixture package. The nominal composition of the inhibitor package shall be 100 percent diethylthiourea. Allowance is made for the inclusion of small amounts of silica at the expense of the sulfamic acid content specified in 3.1.1.

3.1.1 Sulfamic acid. The average composition for an inspection lot shall range from 91.1 to 95.3 percent sulfamic acid, when tested as specified in 4.3.1. No single sample in a lot shall contain less than 90.0 percent nor more than 96.8 percent sulfamic acid.

3.1.2 Citric acid, anhydrous. The average composition for an inspection lot shall be 5.3 to 7.5 percent anhydrous citric acid, when tested as specified in 4.3.2. No single sample in a lot shall contain less than 4.7 percent nor more than 7.5 percent anhydrous citric acid.

3.1.3 1,3 Diethylthiourea, $C_2H_5NHCSNHC_2H_5$. The acid inhibitor when tested as specified in 4.3.5 shall contain a minimum of 97.0 percent 1,3 diethylthiourea.

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3.1.4 Silica. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2). The acid mixture shall contain microfine silica of average particle size not to exceed 25 nanometers (nm). The silica shall be contained in (blended with) the sulfamic acid. The average composition for an inspection lot shall be 0.10 to 0.20 percent SiO_2 (silica), when tested as specified in 4.3.3. No single sample in a lot shall contain less than 0.05 percent nor more than 0.25 percent silica.

3.1.5 Total acidity (expressed as sulfamic acid). The average total acidity for an inspection lot shall be 102.5 to 103.4 percent, when tested as specified in 4.3.4. No single sample in a lot shall contain less than 101.4 percent nor more than 103.9 percent total acidity.

3.2 Toxic formulations. This material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazard Communication. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify and supplement FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Toxic formulations. The contractor shall have the toxicological formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

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4.2 Sampling for quality conformance inspection. Unless otherwise specified, sampling for quality conformance shall be performed in accordance with MIL-STD-105. For purposes of sampling, an inspection lot for examination and tests shall consist of all material submitted for delivery at one time.

4.2.1 Examination of filled containers. A random sample of filled containers, taken from each lot in accordance with MIL-STD-105 at inspection level I and acceptable quality level 2.5 percent defective, shall be examined to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests. Samples shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan in accordance with MIL-STD-105 the lot represented by the sample shall be rejected.

4.2.2 Testing of the end item. When specified in the contract or order, a certificate of compliance and test reports shall be prepared (see 6.2.2). The end item shall be tested for the applicable characteristics as specified in table I for each lot presented for inspection. The sample unit of the acid mixture shall be a 450-gram (1-pound) sample. The sample unit of the acid inhibitor shall be a 110 gram (1/4-pound) sample. Five sample units of the acid mixture and of the acid inhibitor shall be randomly selected from the lot. The acid mixture sample units shall be combined to form a 2.3 kilogram (kg) (5-pound) composite for tests. The acid inhibitor sample units shall be combined to form a 560-gram (1-1/4-pound) composite for tests. Two or more determinations shall be performed on each composite with no evidence of failure to meet the specified requirements. The lot size shall be expressed in units of containers of the scale removing agent.

TABLE I. Instructions for testing.

Characteristic	Specification reference		Number of determinations per unit	Results reported numerically to nearest 1/
	Requirement	Test method		
Sulfamic acid	3.1.1	4.3.1	Avg. of 2	0.1 percent
Citric acid, anhydrous	3.1.2	4.3.2	Avg. of 2	0.1 percent
1,3 Diethylthiourea	3.1.3	4.3.5	Avg. of 2	0.1 percent
Silica (acid mixture)	3.1.4	4.3.3	Avg. of 2	0.01 percent
Total acidity	3.1.5	4.3.4	Avg. of 2	0.1 percent

1/ Test reports shall include all values on which average results are based.

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4.3 Test procedures. The acid mixture sample units and the acid inhibitor sample units shall be tested by the following procedures in accordance with table I.

4.3.1 Sulfamic acid.

4.3.1.1 Reagents.

(a) Standard sodium nitrite solution 0.25 normal (N). Prepare a solution which contains 17.25 grams sodium nitrite per liter (L). Standardize against purified sulfamic acid.
(b) The sulfamic acid shall be purified according to the following procedure: Dissolve 125 grams of sulfamic acid in 300 milliliters (mL) of distilled water previously heated to 70 degrees Celsius (°C). Filter three times using suction as the solution temperature is falling. Discard any insoluble crystalline material formed in the solution. Cool the filtrate rapidly in an ice-salt bath and let stand for 20 minutes. Wash the crystals with a small portion of ice water, then twice with ethyl alcohol, and finally with ether. Air dry for 1 hour, grind in an agate mortar and store in a desiccator.
(c) Sulfuric acid - 50 percent by volume.
(d) Potassium bromide - saturated solution.

4.3.1.2 Apparatus.

(a) pH meter with millivolt (mV) and polarizing current capability.
(b) Silver billet electrode.
(c) Platinum electrode.
(d) Buret - fifty mL capacity graduated in 0.1 mL divisions, with deliveries controlled to better than 0.05 mL.
(e) Volumetric flask - 1000 mL capacity.
(f) Magnetic stirrer.
(g) Magnetic stirring bar.
(h) Pipet - 100 mL delivery.
(i) Beaker - 400 mL capacity.
(j) Ice-water bath.

4.3.1.3 Test method. Dissolve a 10-gram sample of the acid mixture (weighed to the nearest milligram (mg)) in water in a 1000-mL volumetric flask. Dilute to the mark and mix. Use cool water, 25°C or less, throughout. Transfer a 100 mL aliquot of the freshly prepared solution to a 400 mL beaker. Add 15 mL of the potassium bromide solution. Cool to 15°C in an ice-water bath. Cool the sulfuric acid solution in the ice-water bath. Connect the silver billet electrode with a polarizing jumper to the input jack of the pH meter. Connect the platinum electrode to the reference jack. Set the meter to read millivolts. Connect meter to power supply. Add 25 mL of the sulfuric acid solution to the sample solution. Agitate the solution by means of the magnetic stirrer and stirring bar. Immerse the active surfaces of the electrodes in the solution. Titrate the solution with standard 0.25N sodium nitrite such that the buret delivery tip is just below the surface of the sample solution. The equivalence point shall be taken as the point at which further additions of the titrant will not cause any additional surge in the galvanometer. Calculate as follows:

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$$\text{Percent sulfamic acid} = 9.701 \frac{NV}{W}$$

Where:

N = Normality of standard sodium nitrite solution.
V = Volume of standard sodium nitrite solution in mL.
W = Grams of sample in the aliquot.

4.3.2 Citric acid.

4.3.2.1 Reagents.

(a) Potassium permanganate solution - Dissolve 5 grams $KMnO_4$ in distilled water and dilute to 100 mL.
(b) Ferrous sulfate solution - Dissolve 200 grams $FeSO_4 \cdot 7H_2O$ in distilled water. Dilute to 500 mL and add 10 mL of 1:1 H_2SO_4.
(c) Potassium bromide crystals.
(d) Sulfuric acid - 50 percent by volume.
(e) Sodium sulfate - anhydrous powder.
(f) Ethyl alcohol - 190 proof.
(g) Ethyl ether - anhydrous.

4.3.2.2 Apparatus.

(a) Pipet - 100 mL delivery.
(b) Buret - two 50 mL burets.
(c) Cylinder - glass, 5 or 10 mL capacity.
(d) Volumetric flask - 500 mL capacity.
(e) Volumetric flask - 100 mL capacity.
(f) Erlenmeyer flask - 250 mL capacity, glass stoppered.
(g) Filtering flask - 250 or 500 mL capacity.
(h) Crucible - Gooch, porcelain, coors, 270-3, or equivalent.
(i) Filter paper - 2.1 centimeters (cm), glass fiber no. 934, A.H. Burlbot Paper Company, or equivalent.
(j) Suction pump.
(k) Glass desiccator - containing concentrated H_2SO_4.
(l) Drying oven - 105°C.
(m) Laboratory hood.
(n) Mechanical shaker.
(o) Ice-water bath.

4.3.2.3 Test method. Transfer a 100-mL aliquot from the solution prepared in 4.3.1.3 to a 150-mL glass stoppered Erlenmeyer flask. Add 2 grams KBr and 10 mL of 1:1 H_2SO_4. Shake to dissolve and, if necessary, heat to about 50°C. Let stand 5 minutes. Add 20 mL $KMnO_4$ solution slowly from a buret in 1-2 mL portions in the hood. Swirl the flask a few seconds after each addition. Allow to stand 5 minutes undisturbed. Cool to 15°C. Add 20 mL of ferrous sulfate solution slowly while swirling the flask to dissolve the MnO_2 completely. Add 20 grams Na_2SO_4 (anhydrous) with shaking to dissolve it. Continue shaking for 5 minutes to completely dissolve the Na_2SO_4. Repeat the determination if the Na_2SO_4 remains undissolved. Do not confuse undissolved Na_2SO_4 with the pentabromacetone precipitate. Cool to 15°C and shake vigorously for 5 minutes. Immediately, while still cold, collect the pentabromacetone in the Gooch crucible

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and wash the residual precipitate from the flask with portions of the filtrate. Add 50 mL of cold water to the flask and wash the crucible and contents. Use suction to speed the filtration and drying. Dry the crucible overnight in a desiccator over H_2SO_4 and weigh. Record the weight as A. Place the crucible on the filter flask and dissolve the pentabromacetone with alcohol followed by ether, filling the crucible 3 times with each solvent. Dry the crucible 10 minutes at 105°C in an oven. Cool the desiccator for 30 minutes and reweigh. Record the weight as B. Calculate as follows:

$$\text{Percent citric acid (anhydrous)} = 42.4(A-B)$$

4.3.3 Silica (acid mixture).

4.3.3.1 Reagents.

(a) Hydrofluoric acid - 50 percent solution.
(b) Sulfuric acid - 50 percent solution.

4.3.3.2 Apparatus.

(a) Laboratory hood.
(b) Platinum crucible - 50 mL capacity.
(c) Sand bath.
(d) Desiccator.

4.3.3.3 Test method. Weigh accurately 5.000 ± 0.500 grams (W) of the dry acid mixture into a platinum crucible. Heat carefully in the hood to destroy organic matter. Ignite to constant weight at red heat. Cool in desiccator and weigh (W_1). Moisten the residue with a few drops of water. Add 4-6 drops of 1:1 H_2SO_4 and approximately 10 mL of 50 percent hydrofluoric acid solution. Evaporate, without boiling, in the sand bath until fumes of SO_3 are emitted. Fume to dryness. Finally ignite at red heat for 15 minutes. Cool in desiccator and reweigh (W_2). Calculate as follows:

$$\text{Percent silica} = \frac{100(W_1 - W_2)}{W}$$

4.3.4 Total acidity.

4.3.4.1 Reagents.

(a) Standard sodium hydroxide - 0.2N.
(b) Phenolphthalein indicator solution - 1 percent phenolphthalein dissolved in 50 percent alcohol.

4.3.4.2 Apparatus.

(a) Beaker - 250 mL capacity.
(b) Pipet - 50 mL delivery.

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4.3.4.3 Test method. Pipet a 50-mL aliquot of the solution prepared for 4.3.1.3 into a 250 mL beaker. Add a few drops of phenolphthalein indicator. Titrate with the 0.2N sodium hydroxide until a faint pink color persists. Calculate as follows:

$$\text{Percent total acidity (as sulfamic acid)} = 9.701 \frac{VN}{W}$$

Where:

V = Volume of standard sodium hydroxide solution in mL.
N = Normality of the standard sodium hydroxide solution.
W = Weight in grams of sample in the aliquot.

4.3.5 Diethylthiourea.

4.3.5.1 Reagent.

(a) Diethylthiourea - pure.

4.3.5.2 Apparatus.

(a) Pipet - 5 mL delivery.
(b) Pipet - 10 mL delivery.
(c) Volumetric flasks (2) - 100 mL capacity.
(d) Volumetric flask - 500 mL capacity.
(e) Matched silica absorption cells - 10 millimeter path.
(f) Spectrophotometer - Beckman model DU, or equivalent.

4.3.5.3 Test method. Dissolve an accurately weighed 0.25-gram sample of the acid inhibitor in water. Transfer to a 500-mL volumetric flask; dilute to the mark and mix. Pipet 10 mL into a 100-mL volumetric flask. Dilute to the mark and mix. Pipet 5 mL of the second dilution into a 100-mL volumetric flask. Dilute to the mark and mix. Measure the absorbance of the final solution at 235 nm. At the same time, measure the absorbance of standard solutions prepared from accurately weighed 0.20, 0.25 and 0.30-gram samples of pure diethylthiourea dissolved and diluted in the same manner as the acid inhibitor sample. Prepare a calibration curve plotting absorbance versus weight (grams) of standard sample per 100-mL final dilution. Determine the diethylthiourea content of the final sample dilution by comparing its absorbance with the calibration curve. Caculate as follows:

$$\text{Percent Diethylthiourea} = 100 \frac{W_2}{W_1}$$

Where:

W_1 = Weight in grams of sample in final dilution.
W_2 = Weight in grams of diethylthiourea in final dilution sample.

4.4 Inspection of packaging. Sample packs, and the inspection of the packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packing. Packing shall be level B or C as specified (see 6.2.1).

5.1.1 Level B. The scale removing compound shall be packed in 45.4-kg (100-pound) quantities of the agent in fiber drums in accordance with type III, grade A of PPP-D-723. Drums shall be provided with a 50-micrometer (µm) (2-mil) minimum polyethylene liner containing 43.1 kg (95 pounds) of the acid mixture and a 50-µm (2-mil) minimum polyethylene bag containing 2.3 kg (5 pounds) of acid inhibitor.

5.1.2 Level C. The scale removing compound shall be packed in 45.4-kg (100-pound) quantities of the agent in fiber drums in accordance with type I, grade A of PPP-D-723. Drums shall be provided with a 50-µm (2-mil) minimum polyethylene liner containing 43.1 kg (95 pounds) of the acid mixture and 50 µm (2-mil) minimum polyethylene bag containing 2.3 kg (5 pounds) of acid inhibitor.

5.2 Marking. In addition to any special marking specified in the contract or order, shipments shall be labeled and marked in accordance with MIL-STD-129 and with 29 CFR, Part 1910.1200, and shall include the date of manufacture. The following precautionary note shall be included.

"WARNING: Do not mix scale removing compound with boiler passivator compound, nitrates, or strong oxidizing agents. Fumes and fire may result. Do not breathe fumes. Wash down thoroughly if mixed accidentally. Compound shall be disposed of in accordance with local, state and federal regulations."

6. NOTES

6.1 Intended use. The scale removing compound covered by this specification is suitable for removing scale and sludge deposits normally found in medium and high pressure marine boilers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Additional marking, if required (see 5.2).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.1.4, 4.2.2	Certificate of compliance	DI-E-2121	----
4.2.2	Test report	DI-T-2072	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313 and 29 CFR 1910.1200. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.4 Subject term (key word) listing.

Citric acid
Diethylthiourea
Scale remover
Silica microfine
Sulfamic acid

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-24155B(SH)	BOILER SCALE REMOVING COMPOUND

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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